

Code Deliverable Summary: Fraud Detection Repository

This document summarizes the code ZIP contents and how each file maps to the challenge tasks.

Main command sequence

```
pip install -r requirements.txt
python scripts/run_task1.py
python scripts/train_models.py --dataset both
python scripts/run_shap.py --dataset fraud
python scripts/run_shap.py --dataset creditcard
```

Core modules

File	Purpose
src/data_utils.py	Load and validate schemas; clean duplicates; compute class distributions.
src/features.py	Convert IPs, attach countries, engineer temporal, velocity, and monetary features.
src/preprocessing.py	Build leakage-safe split, preprocessing, and SMOTE flow.
src/visualization.py	Generate required EDA plots.
src/modeling.py	Train Logistic Regression and Random Forest; evaluate AUC-PR, F1, precision, recall, and confusion matrix.
src/explainability.py	Generate SHAP summary plots for the best saved tree model.

Task mapping

Challenge task	Implementation
Task 1	scripts/run_task1.py plus data_utils, features, preprocessing, and visualization modules.
Task 2	scripts/train_models.py and src/modeling.py.
Task 3	scripts/run_shap.py and src/explainability.py.
Testing	tests/test_features.py and GitHub Actions workflow.

Important correctness decisions

Resampling is placed after the stratified split and inside the imbalanced-learn pipeline to avoid leakage and keep the test set representative of real fraud prevalence. AUC-PR and F1-score are prioritized because both datasets are highly imbalanced. The e-commerce and credit card datasets are processed independently, as required by the assignment.